

Practice Questions for 2024-11-07-NetSimplex

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1 Questions

1. In the context of the Network Simplex Method, what is the significance of a spanning tree in representing the network flow problem, and how does it relate to the simplex method's basis?
 - a. A spanning tree is an arbitrary subgraph used for visualization purposes, unrelated to the simplex method's basis.
 - b. A spanning tree represents a feasible solution, and its arcs correspond to the basic variables in the simplex method's basis.
 - c. A spanning tree is a cycle in the network, representing a non-feasible solution, and its arcs are non-basic variables.
 - d. A spanning tree is only relevant in the dual simplex method, not the primal.
 - e. A spanning tree is a disconnected subgraph, and its arcs are not directly related to the simplex method's basis.
2. Why might the initial primal flow obtained by setting flows on arcs outside the spanning tree to zero be infeasible in the Network Simplex Method?
 - a. The spanning tree might not be connected, leading to infeasibility.
 - b. The initial spanning tree might contain cycles, causing infeasibility.
 - c. Some flow variables corresponding to arcs within the spanning tree might have negative values.
 - d. The supplies and demands at the nodes might not be balanced.
 - e. The costs associated with the arcs might be negative.
3. In the Dual Network Simplex Method, what is the significance of the condition that the reduced costs (z_{ij}) for arcs not in the spanning tree are all non-negative?
 - a. This condition guarantees primal feasibility.
 - b. This condition guarantees optimality.
 - c. This condition guarantees dual feasibility.
 - d. This condition is only relevant for the primal simplex method.
 - e. This condition is irrelevant to the feasibility or optimality of the solution.
4. Explain the key difference in how the primal and dual network simplex methods handle the selection of entering and leaving arcs during an iteration.

- a. Both methods use the same rules for selecting entering and leaving arcs.
 - b. The primal method selects the entering arc based on reduced cost and the leaving arc based on the minimum flow in a cycle; the dual method selects the leaving arc based on negative flow and the entering arc based on minimum dual slack.
 - c. The primal method focuses on improving primal feasibility, while the dual method focuses on improving dual feasibility, but the arc selection rules are identical.
 - d. The dual method does not use the concept of a spanning tree, unlike the primal method.
 - e. The entering and leaving arc selection is random in both methods.
5. In Slide 2, an initial primal flow is obtained by setting $x_{ij} = 0$ for $(i, j) \in \mathcal{A} \setminus T$. What is a potential issue with this approach?
- a. The resulting flow might not be optimal.
 - b. The spanning tree T might not be connected.
 - c. The resulting flow might contain negative values.
 - d. The basis matrix B might be singular.
 - e. The vector $\tilde{b}$ might not be consistent.
6. According to Slide 3, what equation defines the relationship between dual variables (y_i, y_j) , reduced costs (z_{ij}) , and arc costs (c_{ij}) for an arbitrary arc (i, j) in the network?
- a. $y_i + y_j + z_{ij} = c_{ij}$
 - b. $y_j - y_i + z_{ij} = c_{ij}$
 - c. $y_i - y_j + z_{ij} = c_{ij}$
 - d. $y_i + y_j - z_{ij} = c_{ij}$
 - e. $y_j - y_i - z_{ij} = c_{ij}$
7. Slide 4 states that adding one arc to a spanning tree T results in what?
- a. A disconnected graph
 - b. A spanning tree with increased cost
 - c. A cycle
 - d. A graph with reduced connectivity
 - e. A minimum spanning tree
8. In Slide 5, what does the presence of red arcs in the network flow diagram indicate?
- a. Arcs with zero flow
 - b. Arcs with positive reduced costs ($z_{ij} > 0$)
 - c. Arcs with negative reduced costs ($z_{ij} < 0$)
 - d. Arcs in the current spanning tree

e. Arcs not in the current spanning tree

2 Answers

1. In the context of the Network Simplex Method, what is the significance of a spanning tree in representing the network flow problem, and how does it relate to the simplex method's basis?
 - a. **Incorrect.** The spanning tree is not arbitrary; it's fundamental to the algorithm's efficiency and directly relates to the simplex method's basis.
 - b. A spanning tree is crucial because it defines a basis for the simplex method. Each arc in the spanning tree corresponds to a basic variable in the linear program representing the network flow. The flow on these arcs determines the solution, while arcs not in the spanning tree have zero flow (non-basic variables).
 - c. **Incorrect.** A spanning tree is acyclic (contains no cycles) and represents a feasible (though not necessarily optimal) solution. Cycles are formed when adding arcs to the spanning tree.
 - d. **Incorrect.** Spanning trees are fundamental to both primal and dual network simplex methods.
 - e. **Incorrect.** A spanning tree is a connected subgraph, and its arcs are directly related to the basic variables in the simplex tableau.
2. Why might the initial primal flow obtained by setting flows on arcs outside the spanning tree to zero be infeasible in the Network Simplex Method?
 - a. **Incorrect.** A spanning tree, by definition, is connected.
 - b. **Incorrect.** A spanning tree, by definition, is acyclic (contains no cycles).
 - c. The initial solution sets flows on arcs outside the spanning tree to zero. However, solving for the flows within the spanning tree might result in some flows having negative values, violating the non-negativity constraints of the problem.
 - d. **Incorrect.** While unbalanced supplies and demands would lead to an infeasible problem, the initial solution assumes balanced supplies and demands.
 - e. **Incorrect.** Negative arc costs do not directly cause infeasibility in the initial solution; they affect the optimality but not the feasibility.
3. In the Dual Network Simplex Method, what is the significance of the condition that the reduced costs (z_{ij}) for arcs not in the spanning tree are all non-negative?
 - a. **Incorrect.** Primal feasibility is achieved when all flow variables are non-negative.
 - b. **Incorrect.** Optimality is achieved when all reduced costs are non-negative and primal feasibility is satisfied.
 - c. In the dual simplex method, the non-negativity of the reduced costs for non-basic variables (arcs not in the spanning tree) is the condition for dual feasibility. The algorithm starts with a dual feasible solution and iteratively improves it until primal feasibility is also achieved.
 - d. **Incorrect.** This condition is crucial for dual feasibility in the dual network simplex method.
 - e. **Incorrect.** This condition is fundamental to the dual network simplex method's approach to finding an optimal solution.
4. Explain the key difference in how the primal and dual network simplex methods handle the selection of entering and leaving arcs during an iteration.

- a. **Incorrect.** The selection rules are fundamentally different, reflecting the different approaches to achieving optimality.
 - b. The primal network simplex method starts with a primal feasible solution (possibly with some arcs having zero flow) and improves it by selecting an entering arc with a negative reduced cost. The leaving arc is chosen to maintain a spanning tree structure. The dual network simplex method starts with a dual feasible solution (all reduced costs non-negative) and improves it by selecting a leaving arc with a negative flow. The entering arc is chosen to maintain a spanning tree and dual feasibility.
 - c. **While the methods aim to improve primal and dual feasibility, respectively, the arc selection rules are not identical.**
 - d. **Incorrect.** Both methods utilize the concept of a spanning tree, although they handle it differently.
 - e. **Incorrect.** The selection of entering and leaving arcs is deterministic and follows specific rules in both methods.
5. In Slide 2, an initial primal flow is obtained by setting $x_{ij} = 0$ for $(i, j) \in \mathcal{A} \setminus T$. What is a potential issue with this approach?
- a. **Answer A is incorrect.** While the initial flow might not be optimal, the primary concern highlighted in Slide 2 is the possibility of negative flow values.
 - b. **Answer B is incorrect.** The definition of a spanning tree in Slide 1 ensures that T is connected.
 - c. Answer C is correct. The slide explicitly states that this initial flow might be infeasible because some x_{ij} could be negative.
 - d. **Answer D is incorrect.** The slide doesn't discuss the singularity of B , and it's not the primary concern of the initial flow construction.
 - e. **Answer E is incorrect.** The consistency of $\tilde{b}$ is not directly addressed in the context of this initial flow construction.
6. According to Slide 3, what equation defines the relationship between dual variables (y_i, y_j) , reduced costs (z_{ij}) , and arc costs (c_{ij}) for an arbitrary arc (i, j) in the network?
- a. **Answer A is incorrect.** This equation does not correctly represent the relationship between the dual variables, reduced costs, and arc costs.
 - b. Answer B is correct. This equation is explicitly given and highlighted in a box on Slide 3.
 - c. **Answer C is incorrect.** This equation does not correctly represent the relationship between the dual variables, reduced costs, and arc costs.
 - d. **Answer D is incorrect.** This equation does not correctly represent the relationship between the dual variables, reduced costs, and arc costs.
 - e. **Answer E is incorrect.** This equation does not correctly represent the relationship between the dual variables, reduced costs, and arc costs.
7. Slide 4 states that adding one arc to a spanning tree T results in what?
- a. **Answer A is incorrect.** Adding an arc to a spanning tree does not disconnect the graph; it remains connected.
 - b. **Answer B is incorrect.** While the cost might increase, the key consequence is the creation of a cycle.

- c. Answer C is correct. The slide explicitly states that adding one arc to a spanning tree creates a cycle.
 - d. Answer D is incorrect. Adding an arc to a spanning tree does not reduce connectivity; it remains connected.
 - e. Answer E is incorrect. Adding an arc to a spanning tree does not necessarily result in a minimum spanning tree.
8. In Slide 5, what does the presence of red arcs in the network flow diagram indicate?
- a. Answer A is incorrect. The slide does not indicate that red arcs have zero flow.
 - b. Answer B is incorrect. Red arcs have negative, not positive, reduced costs.
 - c. Answer C is correct. The slide caption explicitly states that red arcs represent arcs with negative reduced costs, indicating non-optimality.
 - d. Answer D is incorrect. The slide does not indicate that red arcs are in the current spanning tree.
 - e. Answer E is incorrect. The slide does not indicate that red arcs are not in the current spanning tree.